

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester Master of Technology (C.E.)

University Examination January 2011

CE705 Advanced Compiler Construction

Date: 07.01.11, Friday

Time: 10.30 a.m to 1.30 p.m

Maximum Marks: 70

Instructions:

1. Attempt all questions.
2. Answers both sections in separate answer books.
3. Figure to the right indicates full marks.
4. Assume suitable data, if necessary and mention it.

SECTION-I

Q-1

- (A) Predictive parser is a special case of RDP. What are the necessary conditions to be met before the construction of predictive parser? 2
- (B) Buffering is the technique to hold the input before it is processed by the parser. Define the role of input buffer in lexical analysis. 3
- (C) Language Translator is a program used to convert the source program to target program. Write difference(s) between Translators like: Compiler, Interpreter and assembler. 2

Q-2

- (A) Error reporting and diagnostic is an important function of a compiler. Explain in detail about error detection and recovery in syntax analysis. 4
- (B) Pass structure of a compiler is one of the important issues along with the efficiency. Write down the pros and cons of reducing number of phases in compiler. 2
- (C) LEX is a Lexical Analyzer Generator tool to perform lexical analysis of a program. Describe the LEX file structure with an example. 6
- (D) Explain the meaning of following terms with respect to shift reduce parsing by giving suitable examples: 2
1. Handle
 2. Viable Prefix

OR

Q-2

- (A) Transforming the grammar into an appropriate form or template for parsing is an important consideration. Write an algorithm to eliminate left recursion from a given grammar. 3
- (B) Write a parser using YACC which recognizes strings of form $a^n b^n$, where $n \geq 1$. 6
- (C) Describe the issues in designing a lexical analyzer. 3
- (D) Consider the grammar 2
- $S \rightarrow SS+ \mid SS^* \mid a$
- Show that the string $aa+a^*$ can be generated by the grammar. Is the grammar ambiguous?

Q-3

- (A) Construct the LALR (1) parsing table for the following grammar. State that whether the given grammar is LALR (1) or not: 8
- $S \rightarrow Aa \mid bAc \mid Bc \mid bBa$
- $A \rightarrow d$
- $B \rightarrow d$
- (B) List the properties of operator grammar? What are the advantages and disadvantages of it? 6
- Explain error recovery in operator precedence grammar.

OR

- Q-3**
- (A) LALR parser handles more programming languages constructs than SLR grammar. Justify the statement. 8
- (B) Prepare LR(1) item set for the following grammar: 6
- $E \rightarrow E + T \mid T$
 $T \rightarrow T F \mid F$
 $F \rightarrow F^* \mid a \mid b$

SECTION-II

- Q-4**
- (A) Generate 3-Address for 4
- $C(A(i,j)) = B(i,j) + C(a(i,j)) + D(i+j)$
- (B) Prepare Syntax Directed Translation Scheme for NOT ($A > B$) OR ($C \leq D$) AND X. 3
- Q-5**
- (A) Find l-value or r-value for the following expressions: 3
1. $a(i+1)$ 2. a 3. $\&a$ 4. $\&(*a)$ 5. $*(&a)$ 6. $*(&(&a))$
- (B) What is code optimization? What are the sources of optimization? Explain with example. 5
- (C) Name at least six items of run-time information stored in an activation record/frame. For each item, identify whether its value is set before the procedure is called, during procedure execution or right before procedure return. 6

OR

- Q-5**
- (A) List the criteria for selecting a code optimization technique. 3
- (B) Explain how address generation instructions are generated by a compiler for dynamic memory. 5
- (C) Symbol table is an abstract data structure associated with a compiler. What is the need for symbol table? What are the attributes of it? Explain tree symbol table in brief. 6
- Q-6**
- (A) Prepare DAG representation for: $Sum = Sum + ((a/b)*c)/(a/b)$. 4
- (B) Why the case that statically allocated languages cannot support recursion and dynamic memory allocation? 4
- (C) For the following C fragment, list the lexemes and corresponding tokens. 6
- ```
main ()
{
 int m, x; /* a simple program for finding *\ max of two*/ numbers*/
 m = 23; x = 4.54; if(x > m) printf ("max is no1");
 else printf ("max is no2");
}
```

OR

- Q-6**
- (A) Write a short note on following: 4
1. loop unwinding
2. Dead Variables and Dead Codes
- (B) Explain how preprocessor, linker and loader support in overall working of the compiler. 4
- (C) Construct a minimal state DFA for the following regular expression: 6
- $(a \mid b)^* \mid (ab)^*b \mid a^*(bb)^*$